Ministry of Education
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Collage of Computer Science and Engineering

CCSW 437 – User Experience (Fall 2026)
Final Design Project Report

Project Title: Gemini – User Experience Study

Usability Testing of Gemini AI Assistant's Task Completion and Frequent Use.

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1.Project Description:

Our project focused on conducting usability testing for the Gemini app, one of Google's newest and rapidly growing AI tools. Given its increasing popularity, we aimed to evaluate how effectively the app performs in completing everyday tasks and how intuitive it is for regular use. The primary objective was to determine whether Gemini provides a truly user-friendly experience or if there are usability challenges that impact interaction. To achieve this, we carried out a pilot study to identify areas where the app lacked efficiency or clarity. This was followed by testing sessions with participants, allowing us to gather direct feedback on usability issues. Based on the insights derived from surveys and analysis, we then redesigned selected interface components to address the problems observed, ensuring that the app's usability could be improved and better aligned with user expectations.

1.1Target Application:

- **Application Name:** Gemini (Google's Artificial Intelligence Application).
- **Platform:** Mobile App Stores & Web Version.
- **Website URL**: Google Gemini

2. Evaluation Goals:

The following research questions and goals guided our evaluation process:

1. Evaluate the overall usability of the Gemini app: Assess how easy and intuitive the app is for users during regular, everyday use.
2. Examine how effectively users can complete common tasks: Identify whether users can accomplish typical tasks efficiently and without confusion.
3. Identify usability issues that may negatively impact user experience: Detect navigation problems, unclear features, or interaction difficulties that affect usability.
4. Assess user understanding and learnability of Gemini's features: Determine how easily users understand and learn how to use both basic and advanced features.

5. Measure user satisfaction with the Gemini app experience: Explore how satisfied users feel overall when interacting with the app.
6. Evaluate whether Gemini meets user expectations as a daily AI tool: Compare user expectations with the actual performance and usefulness of the app in daily life.
7. Explore the reliability and practicality for frequent users: Understand how well the app supports users who rely on AI tools regularly.

3.Targeted Participants:

To ensure a comprehensive evaluation, we selected participants from diverse backgrounds to represent the actual user base of Gemini. Our target groups include:

1. **University Students (Ages 18–30):** Selected because many students rely heavily on AI tools like Gemini for studying, summarizing lectures, academic writing, and research purposes.
2. **Young Professionals (Early Career Employees):** Participants who often use AI to support productivity, streamline communication, solve problems, and assist with daily work tasks.
3. **Everyday Users Interested in AI:** Individuals who use mobile AI tools for general daily tasks, such as planning schedules, brainstorming ideas, or asking quick questions.
4. **Users with Mixed Technical Backgrounds**: This group includes a range of expertise to reveal different usability challenges:
    - **Beginners:** Users with little to no experience using AI.
    - **Experienced Users:** Users familiar with chatbots and digital assistants.
5. **Multilingual Users:** Since Gemini supports multiple languages (including English and Arabic), testing with multilingual participants helps evaluate the app's performance across different language groups.

4. Recruitment Process:

Our recruitment process involved distributing an online survey using Google Forms, which was shared with family members, friends, and personal contacts through messaging applications and social media platforms. This approach was chosen to efficiently collect responses from users who are familiar with mobile and web applications. The target group consisted of university students, young adults, and general users who use AI tools or digital applications in their daily lives. These participants represent typical Gemini users with varying levels of technical experience. No strict selection criteria were applied. Participants were only required to, Have access to a smartphone or web browser, Possess basic familiarity with using mobile or web applications, Be able to understand English or Arabic. Participation was voluntary, and consent was implied by completing the survey. All responses were collected anonymously, and no personal identifying information was requested. Through this recruitment process, we collected 43 valid responses, exceeding the minimum requirement of 10 participants for the study.

5. Study Design:

We conducted a within-subject study to evaluate the user experience effectively (For more details on this methodology, refer to: NNGroup Article). In this approach, each participant first experienced the Gemini app in its original form. Based on the issues identified, we redesigned parts of the interface. Although participants have not yet tested the redesigned version, this study design allows us to later compare their experiences across both versions using the same group.

5.1Why Within-Subject Design?

We chose this methodology for the following reasons:

- **Direct Comparison:** Since the same participants experience both versions, they can directly compare usability and performance, yielding more reliable insights.

- **Reduced Variability:** Individual differences (such as prior experience with AI apps or personal preferences) are controlled because each participant serves as their own baseline.
- **Efficiency:** Fewer participants are needed compared to a between-subject design, while still ensuring meaningful results.
- **Focused Feedback:** Participants can highlight specific improvements or issues they notice across versions, making the redesign evaluation more precise.

6. Evaluation Metrics:

To assess the usability of Gemini, we used the following metrics:

Metric Name	Type of Data	Reason
Task Success	Quantitative (binary: success/failure)	Measures whether participants can complete tasks as intended, showing overall usability and effectiveness of Gemini.
Efficiency (Task Time)	Quantitative (time in seconds/minutes)	Evaluates how quickly users can complete tasks, reflecting ease of use and system responsiveness.
Errors	Quantitative (count of mistakes)	Identifies usability problems by tracking mistakes, misclicks, or failed attempts, highlighting areas needing redesign.
Number of Steps	Quantitative (interaction count)	Shows how complex a task is; fewer steps indicate better efficiency and streamlined design.
Self-Reported Satisfaction (ASQ)	Qualitative (survey responses)	Captures immediate user perception of ease, satisfaction, and frustration after each scenario.
System Usability Scale (SUS)	Standardized survey (quantitative score)	Provides a validated measure of overall usability, allowing comparison with benchmarks across systems.

Net Promoter Score (NPS)	Quantitative (likelihood to recommend)	Indicates user loyalty and long-term acceptance by asking if participants would recommend Gemini to others.
User Satisfaction	Qualitative + quantitative (ratings, comments)	Directly measures how enjoyable and intuitive users find the app, aligning with your goal of testing user-friendliness.

7. Task Scenarios & Data Collection:

The following table outlines the specific tasks participants will perform and the data metrics collected for each task:

Task	Metrics Collected
Find a complex fact & sum it	1. Task Success 2. Efficiency 3. Errors 4. Time on Task
Image generation	1. Task success 2. Self-reported Metrics: After-Scenario Questionnaire (ASQ) for satisfaction 3. Time on Task 4. Errors 5. Efficiency
Find a previous chat	1. Task Success 2. Efficiency 3. Time on Task 4. Errors
Analyze a Document	1. Task Success 2. Errors 3. Efficiency 4. Time on Task
Text translation via Image processing	1. Task Success 2. Time on Task

	3. Errors 4. Efficiency 5. Learnability
Send a Voice Prompt	1. Task Success 2. Time on Task 3. Errors 4. Efficiency
Rate Your Experience	1. User Satisfaction 2. System Usability Scale (SUS) 3. Net Promoter Score (NPS) 4. Time on Task

8. Data Analysis:

To evaluate the user experience of the Gemini application, we analyzed survey responses collected during the pilot study. The survey, completed by 43 participants, was designed with 12 closed-ended questions on a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree) and one open-ended feedback question.

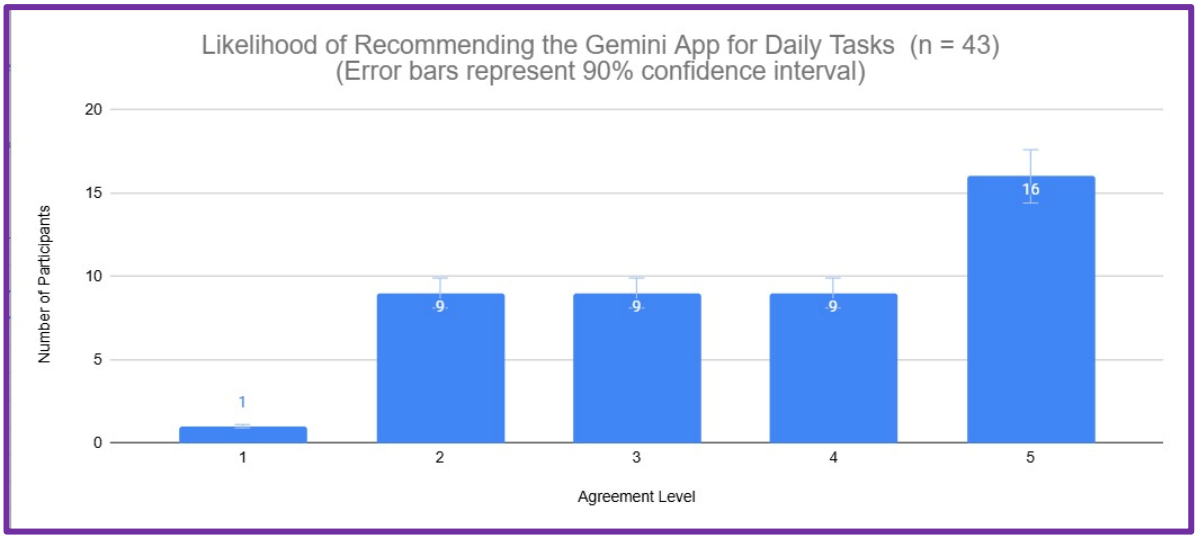
Responses were categorized into four key areas: Usability & Design, Output Quality & Trust, Multimodal Interaction, and Overall Satisfaction. Data was gathered through Google Forms and processed in Google Sheets, where average scores and sentiment trends were calculated. These results were then visualized in figures and linked to the performance metrics from the usability tasks, allowing us to connect survey insights with observed user behavior.

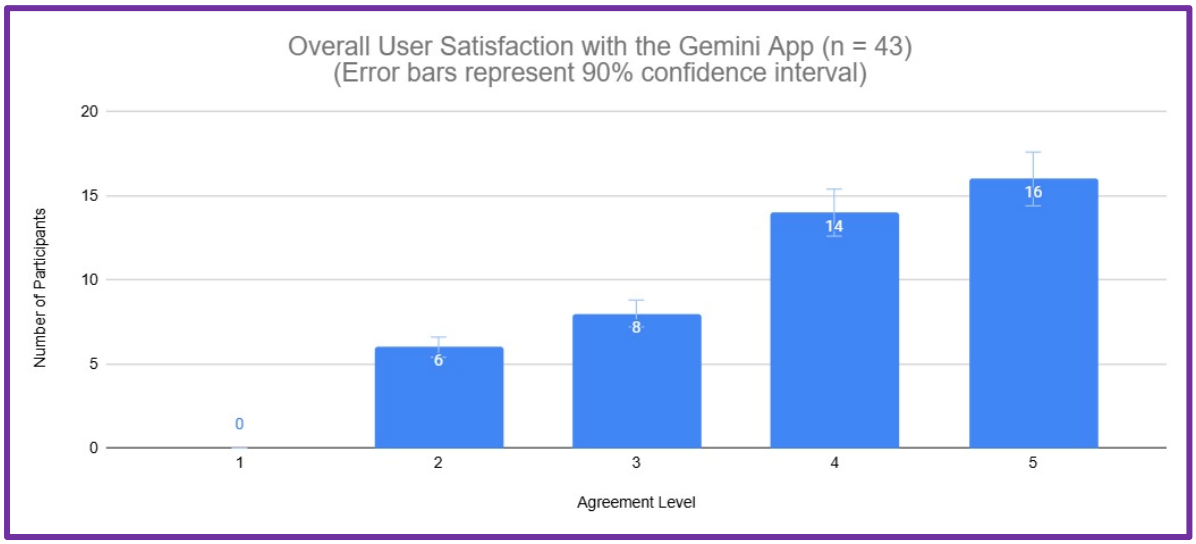
The survey can be accessed via:

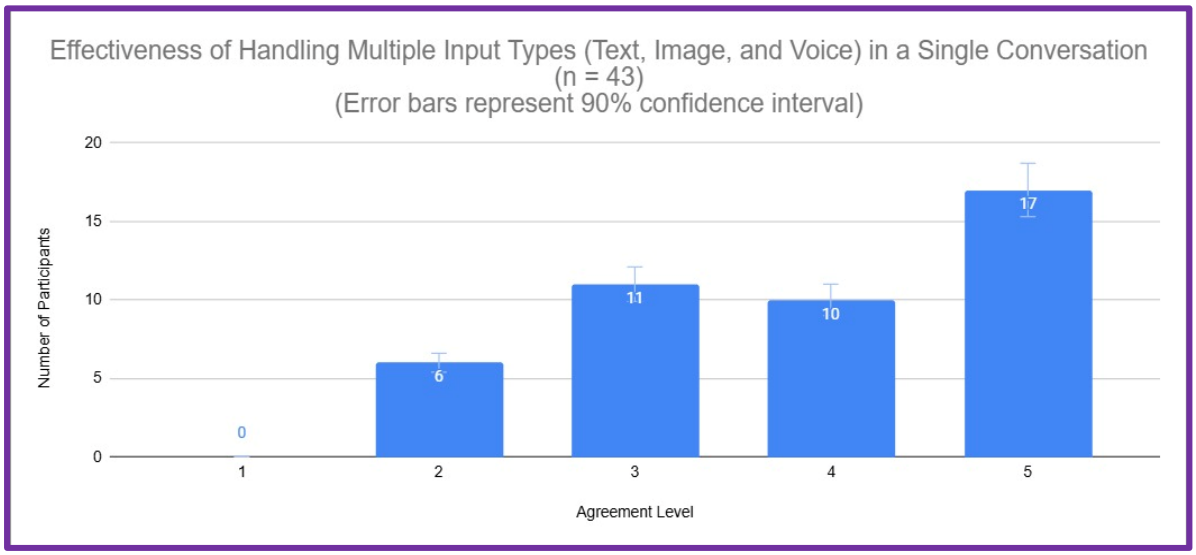
- **Survey Form Link:** https://forms.gle/knQQfYpvYhUMnC9N8
- **Survey Google Sheet Link:** https://docs.google.com/spreadsheets/d/1cJTM3DieRQMDXG5z8cB8ivL1lIoWTGc4SIEOdpGJhfo/edit?usp=sharing

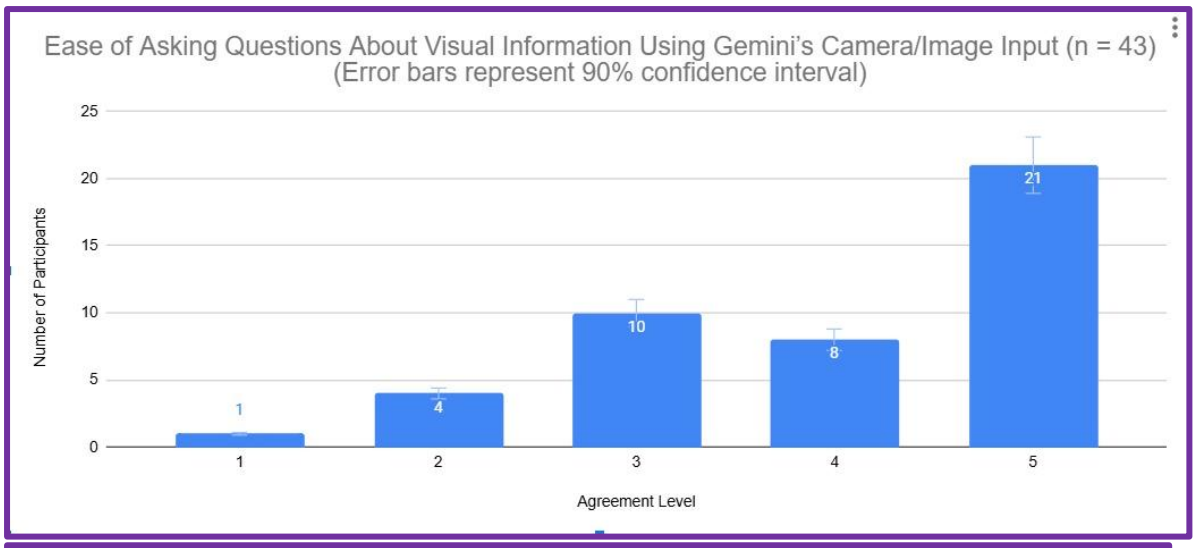
8.1 Survey Analysis Results:

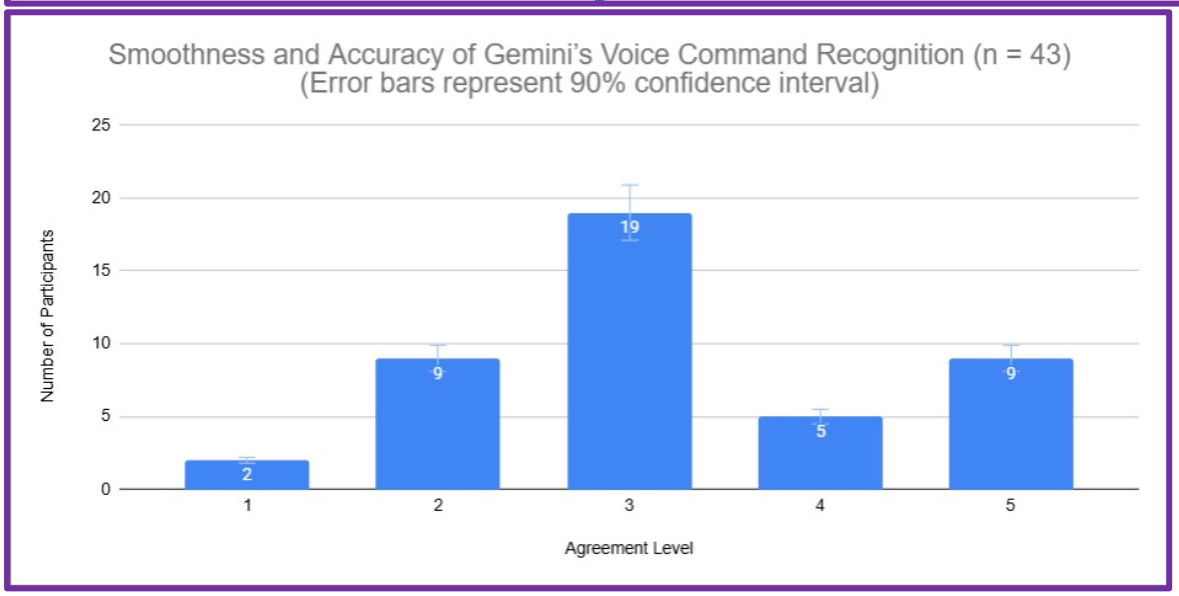
8.1.1 Quantitative Results (Charts):

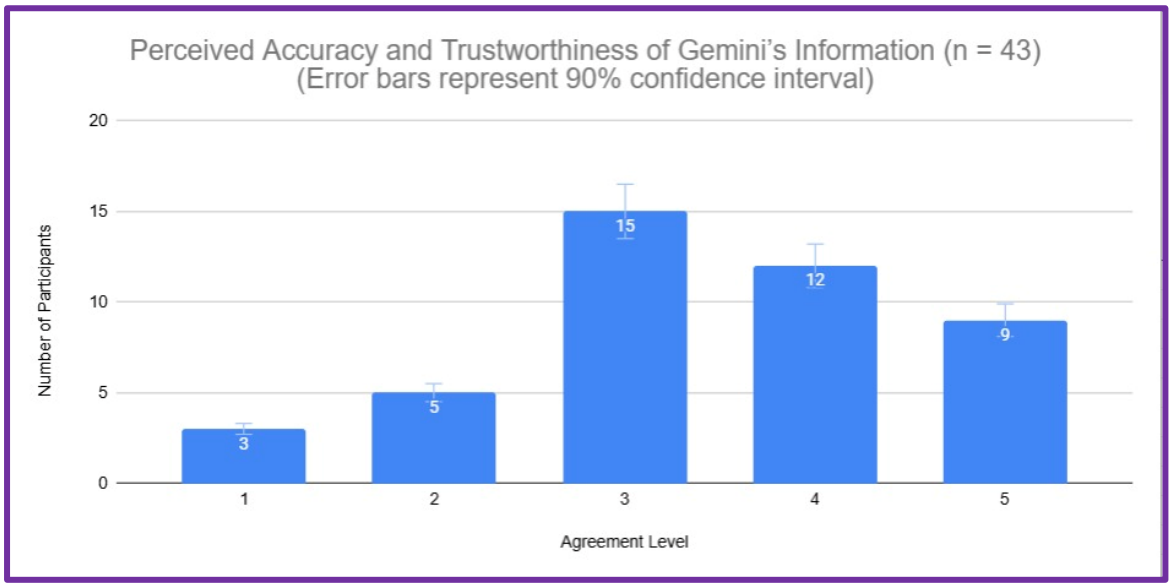


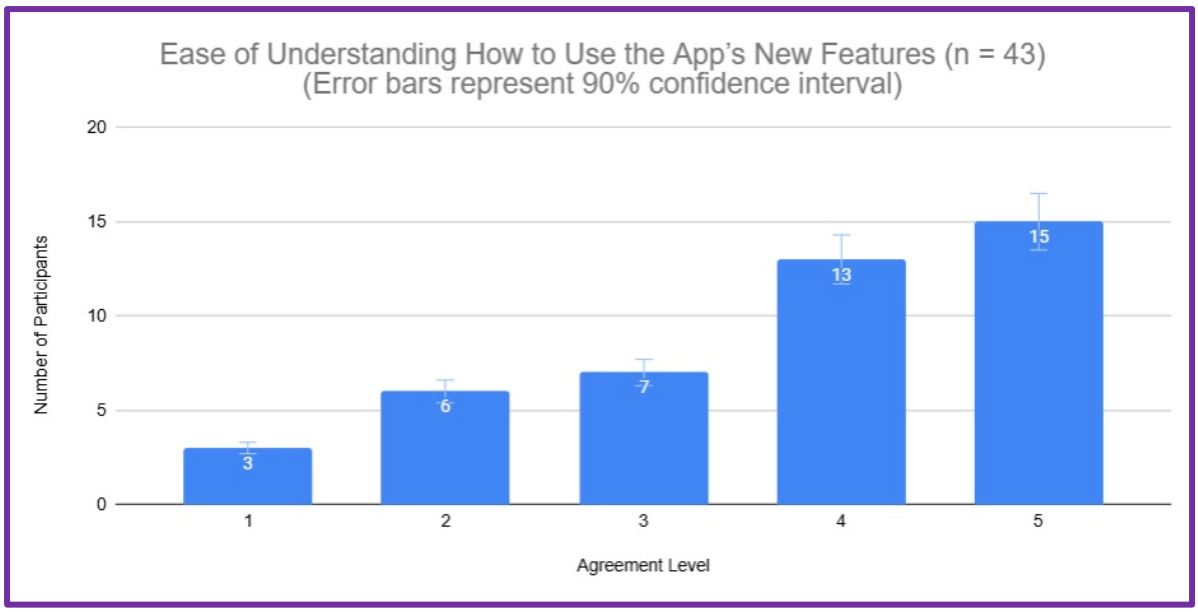


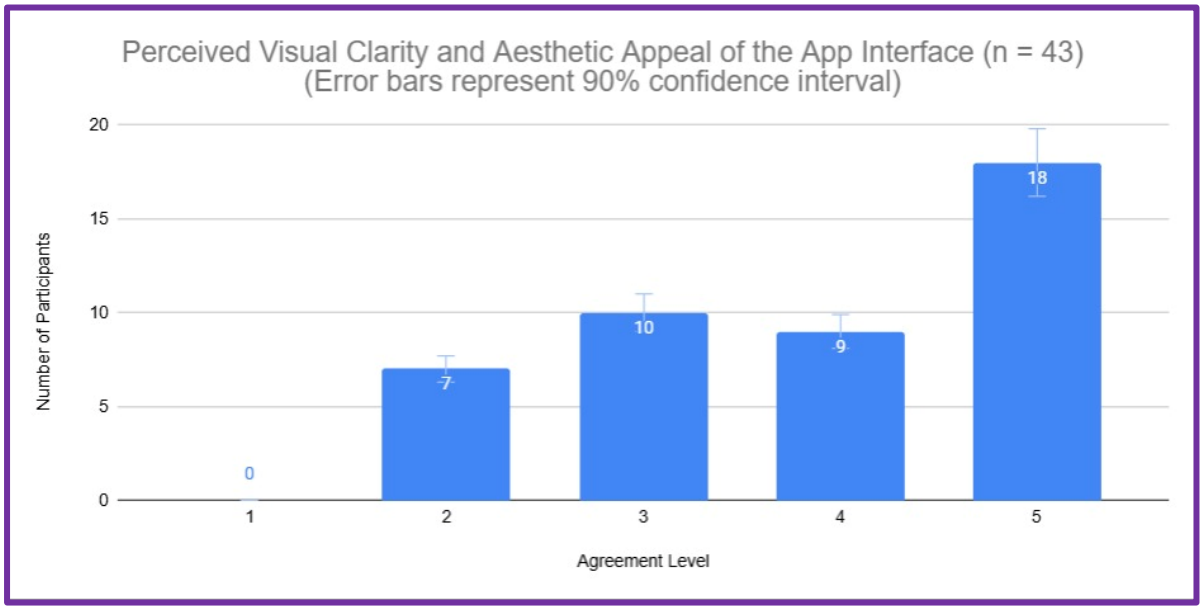


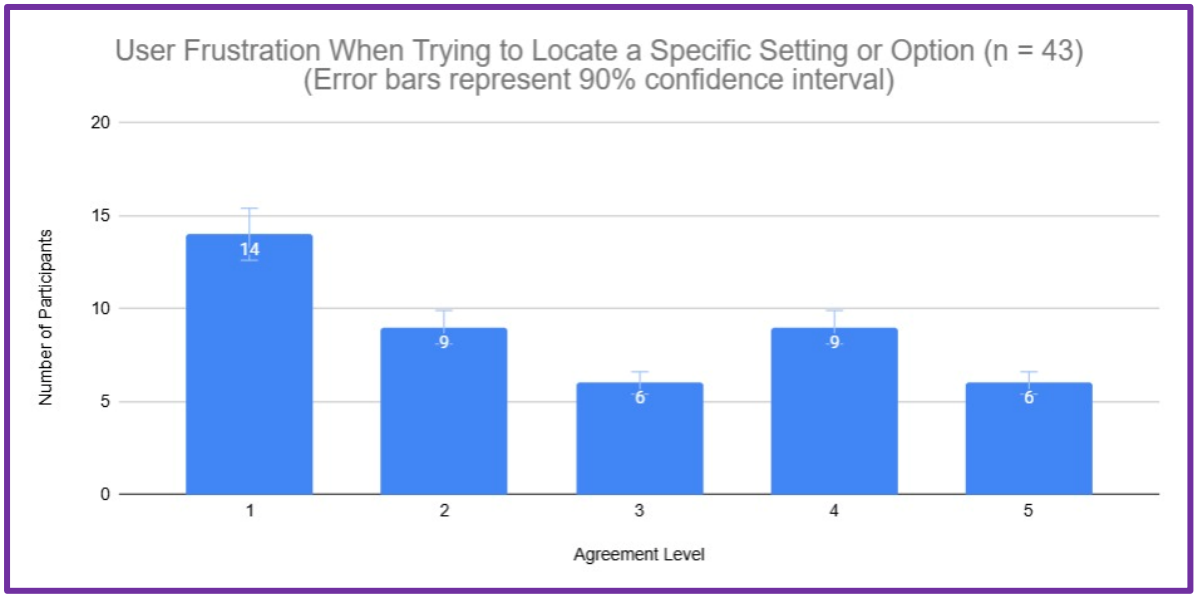

Ease of Asking Questions About Visual Information Using Gemini's Camera/Image Input (n = 43)
(Error bars represent 90% confidence interval)
Number of Participants
Agreement Level
1
4
10
8
21

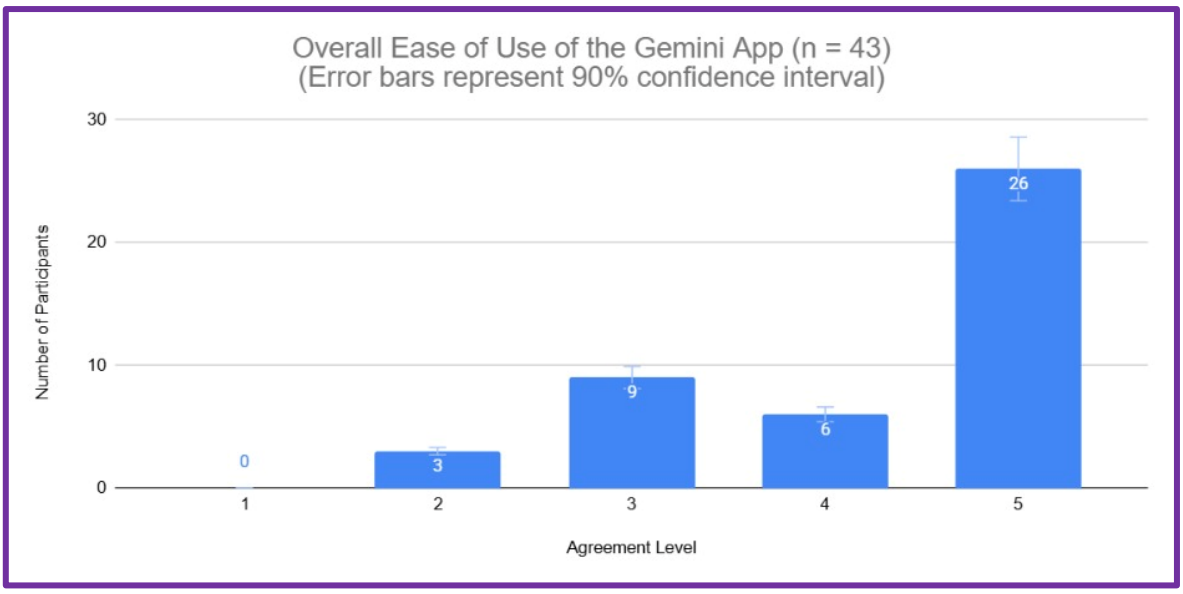

Smoothness and Accuracy of Gemini's Voice Command Recognition (n = 43)
(Error bars represent 90% confidence interval)
Number of Participants
Agreement Level
2
9
19
5
9

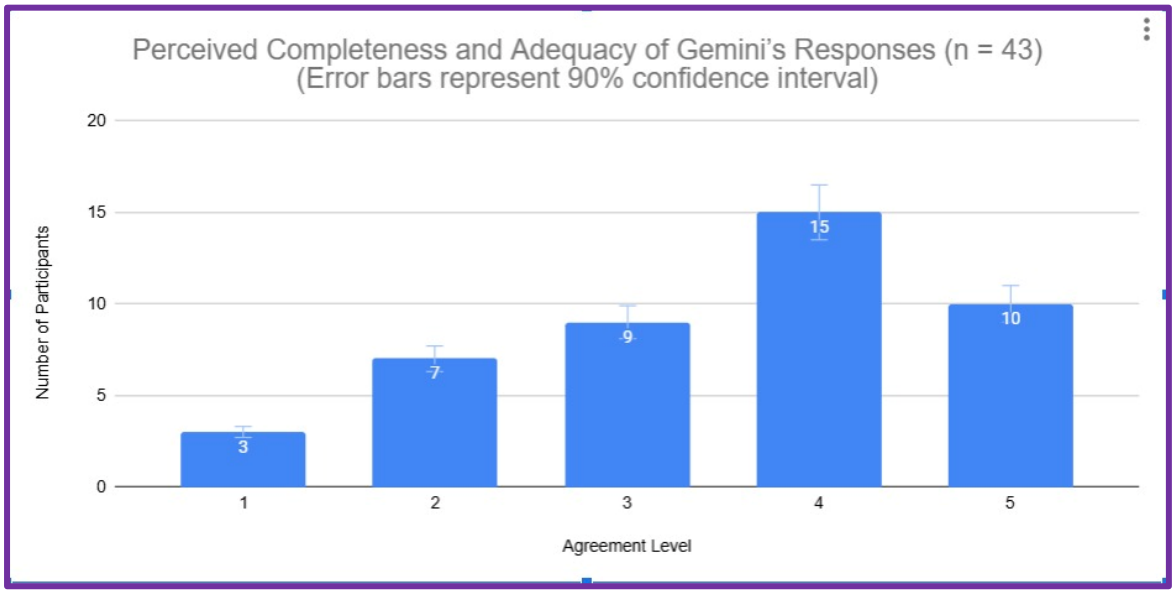

Perceived Accuracy and Trustworthiness of Gemini's Information (n = 43)
(Error bars represent 90% confidence interval)
Number of Participants
Agreement Level
3
5
15
12
9


Ease of Understanding How to Use the App's New Features (n = 43)
(Error bars represent 90% confidence interval)
Number of Participants
20
15
10
5
0
3
6
7
13
15
1
2
3
4
5
Agreement Level


Perceived Visual Clarity and Aesthetic Appeal of the App Interface (n = 43)
(Error bars represent 90% confidence interval)
Number of Participants
20
15
10
5
0
0
7
10
9
18
1
2
3
4
5
Agreement Level


User Frustration When Trying to Locate a Specific Setting or Option (n = 43)
(Error bars represent 90% confidence interval)
Number of Participants
20
15
10
5
0
14
9
6
9
6
1
2
3
4
5
Agreement Level





8.1.2 Qualitative Insights (Open-Ended Feedback):

The open-ended responses provided valuable qualitative insights into areas where Gemini could be improved. Participants highlighted several key themes:

- **Performance & Accuracy:** Participants highlighted the need for better performance and accuracy, particularly faster response times and clearer, more context-aware outputs.

- **Interface & Clarity:** Several users suggested enhancements to the interface and clarity, such as more attractive visuals, improved accessibility of settings, and clearer tool organization.

- **Feature Expansion:** Others expressed interest in new creative features, including video creation, interactive notebook modes, and easier access to image generation.

- **Positive Feedback:** At the same time, a portion of respondents reported positive experiences with no major issues, praising Gemini's integrated design and overall usefulness.

In summary, while the application is generally well-regarded, improvements in speed, output quality, interface design, and feature expansion would further elevate the user experience.

8.2 Metrics Analysis:

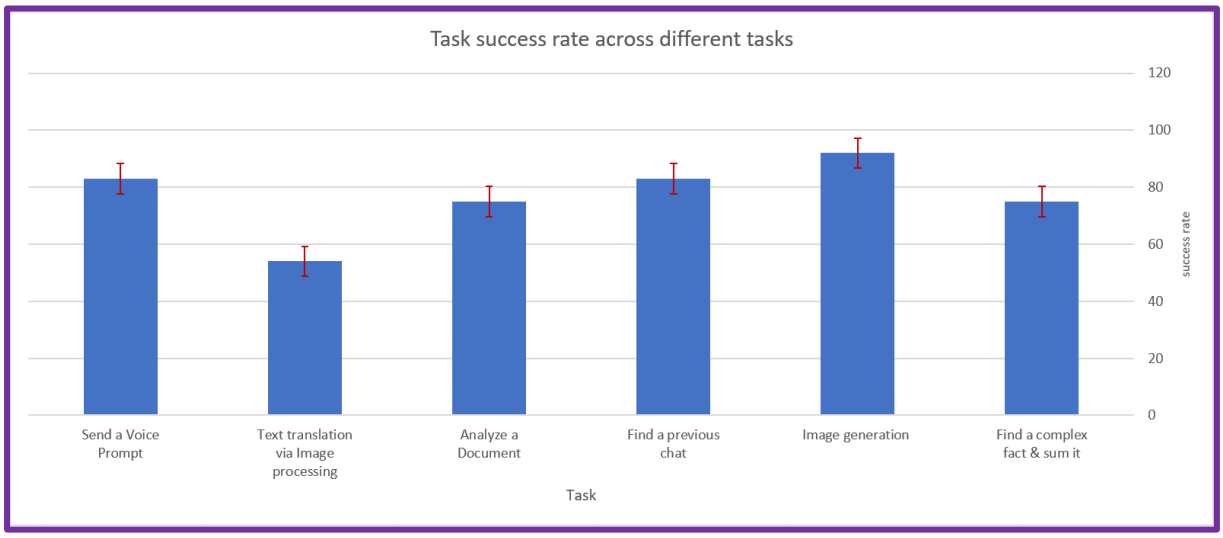
The experiment was conducted by asking participants to perform a series of predefined tasks on the Gemini application. Each task was observed and recorded to capture key usability metrics. Specifically, we analyzed task success, time on task, errors, efficiency, and learnability to evaluate how well participants could complete their goals and how easily they adapted to the system.

These metrics provided both quantitative and qualitative insights into the overall user experience. The figures below illustrate the detailed analysis for each metric across the different tasks.

The sheet could be accessed via :

https://1drv.ms/x/c/3ebe9d5c06ea3d62/IQDlKYmZbF8GQI4B8JcotLDOAfokLdmYioCMjZ11hPN4lsc?e=kLa5cp

8.2.1 Task Success:



Q1 : What is the average success rate for each task?

- Task 1 (Fact & Sum): 75%
- Task 2 (Image Generation): 92%
- Task 3 (Find Chat): 83%
- Task 4 (Analyze Doc): 75%
- Task 5 (Translation): 58%
- Task 6 (Voice Prompt): 83%

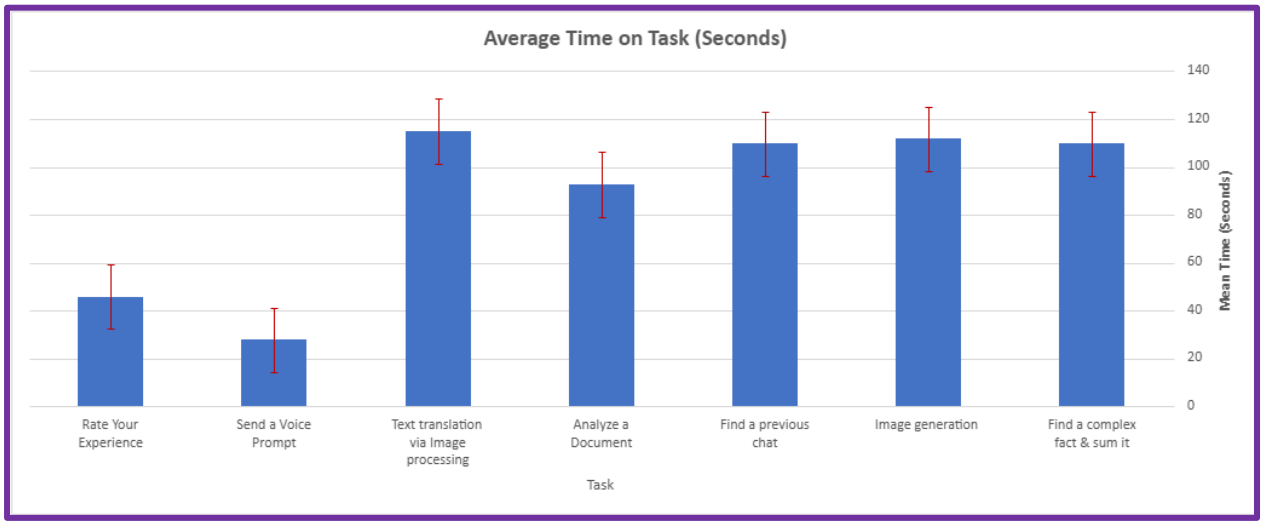
Q2: Which task has the highest success rate?

- Task 2 (Image generation) has the highest success rate at 92%.

Q3: What is the total task success rate of all tasks?

- The total average success rate is 77% (Based on the value 0.766 in cell H14).

8.2.2 Time on task:



Q1: What is the average time spent in each task?

- Task 1 (Fact & Sum): 109.75 seconds
- Task 2 (Image Generation): 111.75 seconds
- Task 3 (Find Chat): 109.75 seconds
- Task 4 (Analyze Doc): 92.67 seconds
- Task 5 (Translation): 114.92 seconds
- Task 6 (Voice Prompt): 27.92 seconds
- Task 7 (Rate Experience): 45.92 seconds

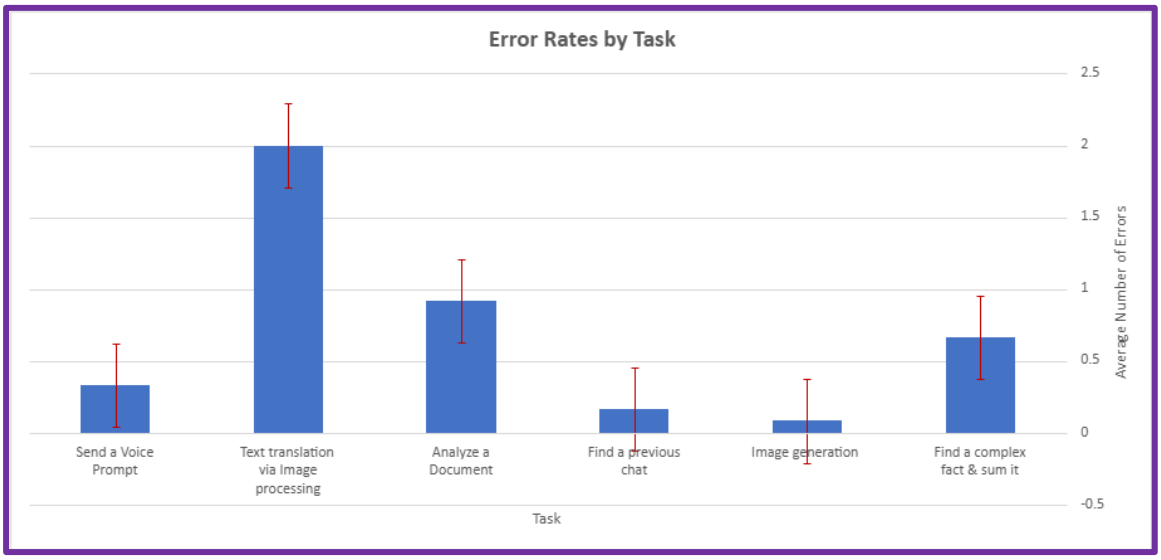
Q2: Which task has the lowest average time?

- Task 6 (Send a Voice Prompt) has the lowest average time at 27.92 seconds.

Q3: What is the average total time on all tasks?

- The overall average time spent on a task is 87.52 seconds (as shown in cell H14).

8.2.3 Errors:



Q1: What is the average number of errors in each task?

- Task 1 (Fact & Sum): 0.67 errors
- Task 2 (Image Generation): 0.08 errors
- Task 3 (Find Chat): 0.17 errors
- Task 4 (Analyze Doc): 0.92 errors
- Task 5 (Translation): 2.0 errors
- Task 6 (Voice Prompt): 0.33 errors

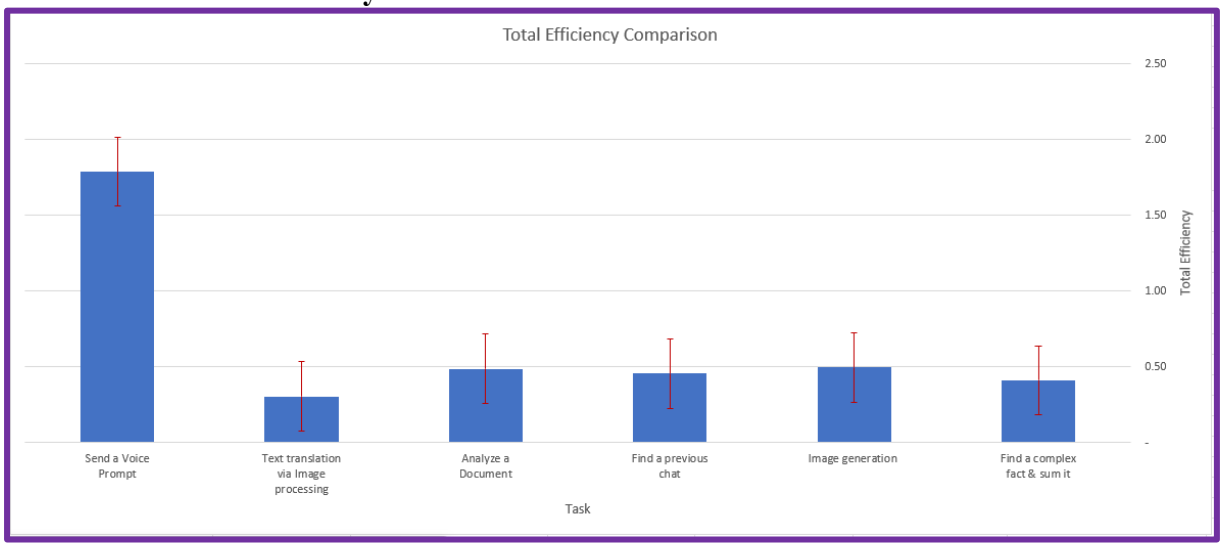
Q2: Which task has the lowest number of errors?

- Task 2 (Image generation) has the lowest average number of errors at 0.08.

Q3: What is the average total number of errors?

- The overall average number of errors across all tasks is 0.70 errors (Calculated as the average of all task error rates).

8.2.5 Efficiency:



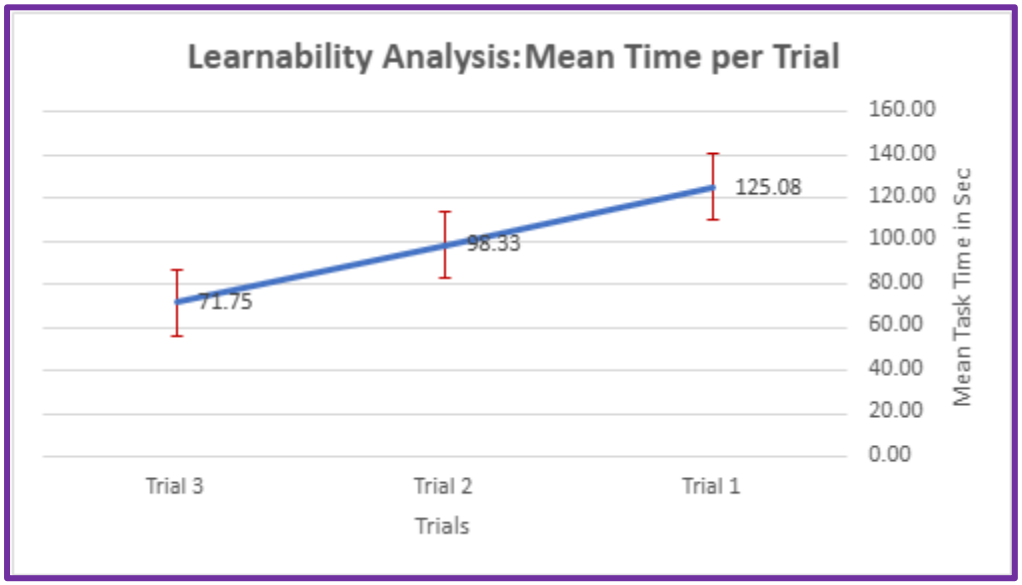
Q1: Based on the three different usability metrics, what is the total efficiency of each task?

- Find a complex fact: **0.41**
- Image generation: **0.49**
- Find a previous chat: **0.45**
- Analyze a Document: **0.49**
- Text translation: **0.30**
- Send a Voice Prompt: **1.78**

Q2: Which task is most efficient?

- Send a Voice Prompt (Task 6) is the most efficient task with a score of 178%. This is due to its high success rate combined with a very short completion time compared to other tasks.

8.2.6 Learnability:



Q1. Is there a learnability effect when comparing the three trials?

Yes, there is a clear learnability effect. The data shows a significant decrease in the average time required to complete the task across the three trials.

- Trial 1 Average: 125.08 seconds (Initial attempt).
- Trial 2 Average: 98.33 seconds.
- Trial 3 Average: 71.75 seconds (Proficiency achieved).

The reduction in time indicates that participants became more efficient and comfortable with the feature as they repeated the task.

9 .UX Design Decisions

Our UX design decisions were directly informed by the usability issues identified during the pilot study and lab testing. After analyzing participant feedback, task performance metrics, and survey results (SUS, ASQ, and NPS), we focused on improving clarity, efficiency, and overall user satisfaction. The following key design decisions were made based on the identified problem areas:

- Improving Interface Clarity and Visual Hierarchy
  Although overall usability scores were positive, a noticeable portion of participants expressed only neutral or slight disagreement regarding interface clarity and visual appeal. To address this, we redesigned the layout with clearer visual hierarchy, better spacing, and more distinguishable sections. Icons and tools were made more visible and logically grouped to reduce cognitive load and make features easier to identify

- Enhancing Feature Discoverability
  Lab testing and some survey responses showed that some users required additional time or steps to locate certain features, or actually did not know how it worked properly, indicating discoverability issues. Based on efficiency metrics (task time and number of steps), we simplified navigation flows and repositioned frequently used features to more prominent locations in the interface, reducing unnecessary interactions.

- Refining Voice Interaction Flow
  While voice commands were generally successful, user feedback rated this feature as mostly neutral, indicating room for improvement. Based on task success rates and user comments, we redesigned the voice interaction flow to provide clearer feedback, confirmation states, and smoother transitions, improving responsiveness and user trust in voice input
  Room for improvement, A backend Aspect rather than User Interfaces :

- Improving Output Confidence and Information Consistency
  Metrics related to Information Quality and Output Completeness received the lowest levels of strong agreement. To respond to this, the redesigned wireframes should emphasize a clearer output structure, better formatting, and visual cues that reinforce reliability, helping users feel more confident in the results provided by the system

- Supporting Accessibility and Ease of Use
  Open-ended feedback highlighted the need for improved accessibility and clearer access to settings. To respond to this, the redesigned wireframes for settings visibility and interaction flows should be better to support users with different experience levels and multilingual backgrounds, aligning with the diverse participant group.

- Introducing User-Requested Features
  Based on qualitative feedback, additions such as a more interactive notebook mode and improved support for image-based tasks are essential. These changes directly respond to user expectations and aim to increase long-term engagement and satisfaction.

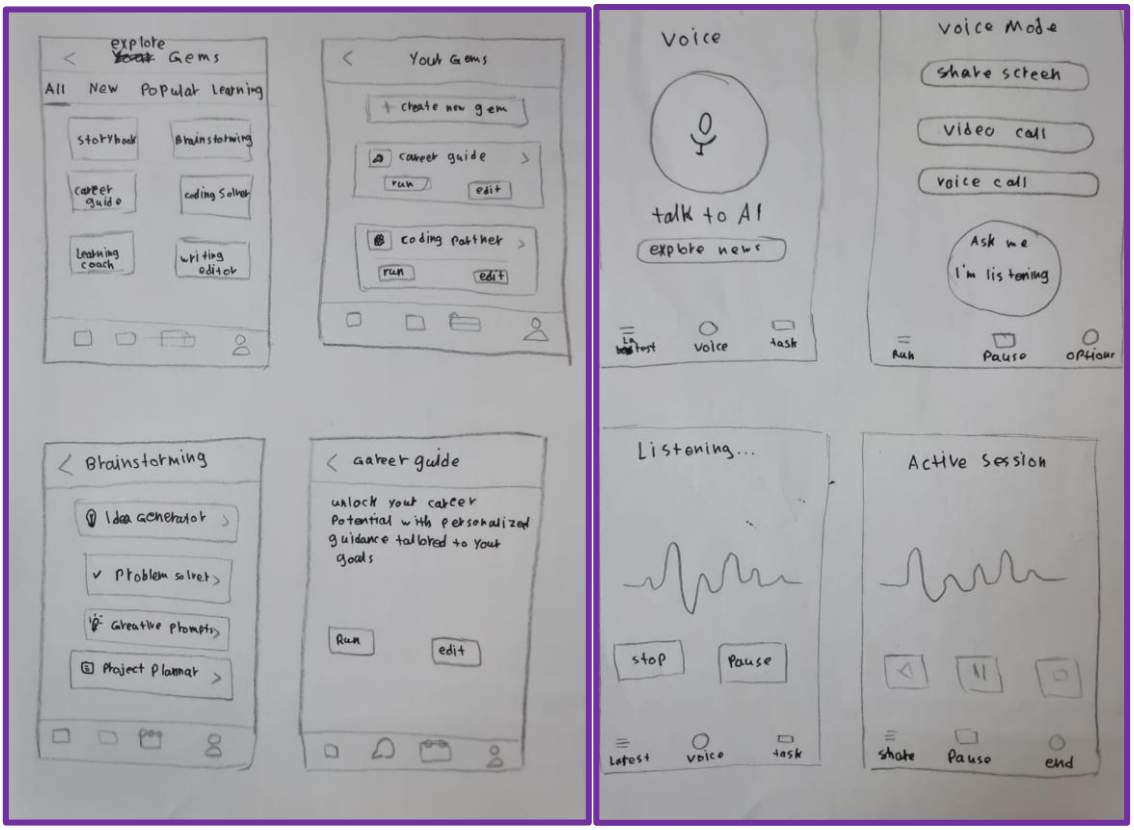
Overall, these design decisions ensure that the redesigned wireframes directly address the usability gaps identified through empirical data, transforming the Gemini experience from generally satisfactory to more efficient, visually clear, and user centered.

10 Wireframe

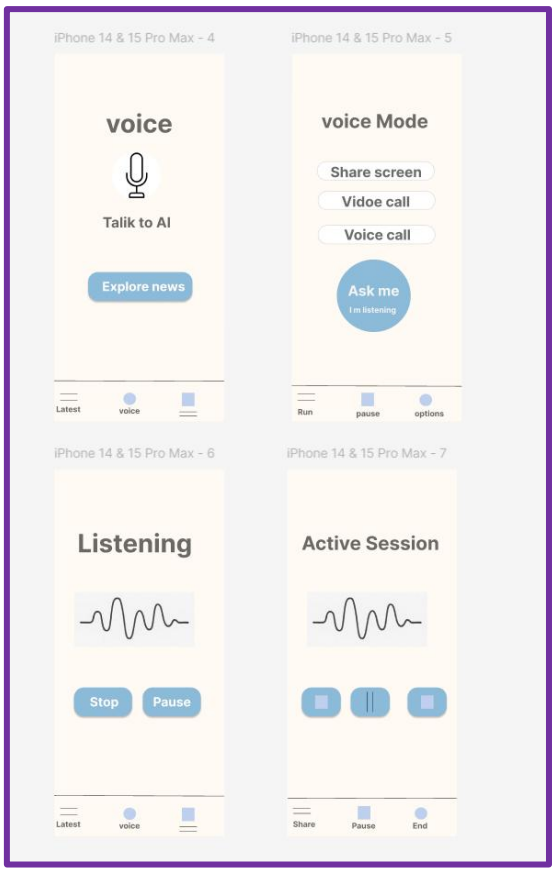
Accessed Via Figma:
https://www.figma.com/design/lVbA1IjIustluWTYy1f0U6/Untitled?node-id=1-67&t=dvu23CmRMdb8MV9D-1

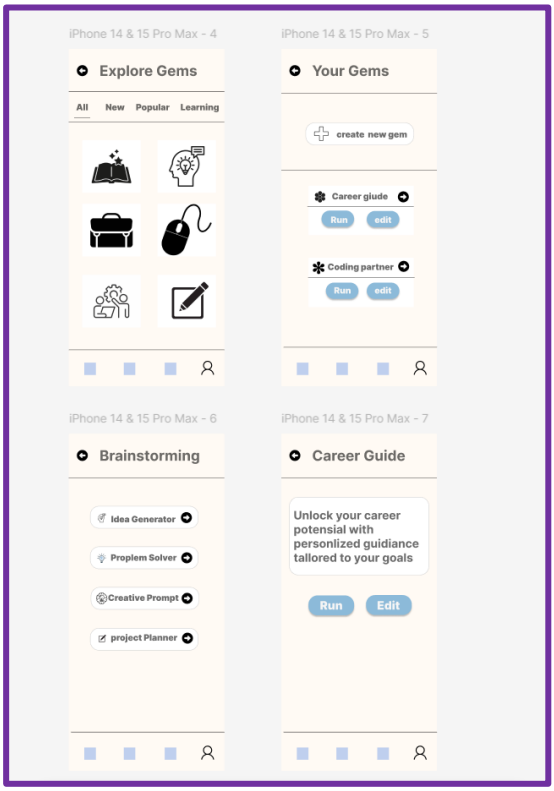
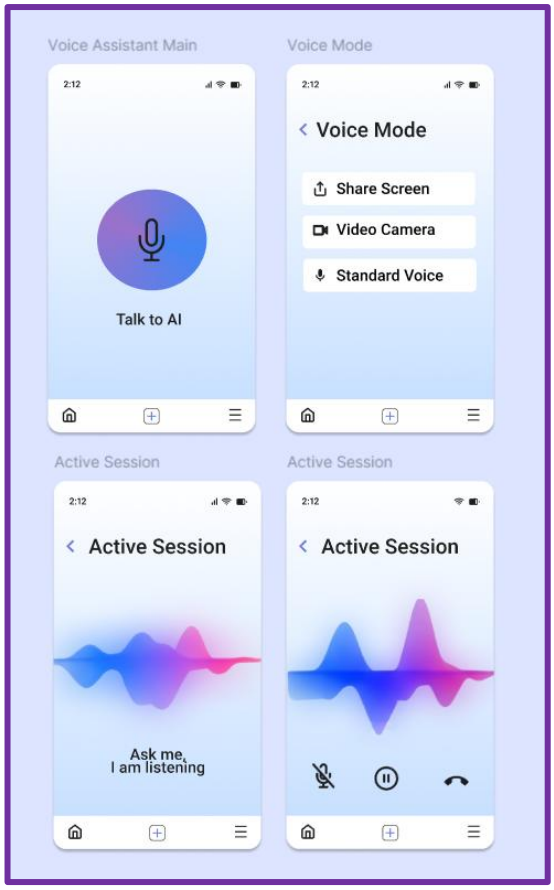
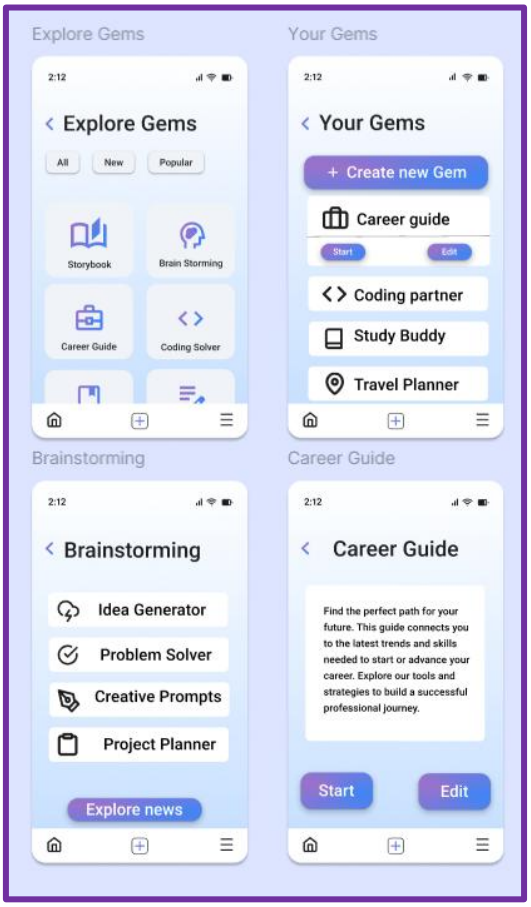
Prototype link : https://www.figma.com/proto/lVbA1IjIustluWTYy1f0U6/Gemini-Wireframe?node-id=106-95&t=Ceufw4tsp17p3rHK-1&scaling=scale-down&content-scaling=fixed&page-id=64%3A2&starting-point-node-id=106%3A95

10.1 Low Fidelity:



10.2 Mid Fidelity:





10.3 High-fidelity Prototype: